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DSA-0310 - CIA-1

Q1) Regular Expressions and DFA

(a) Regular Expressions

1. Register number

^19[0-9]{7}$

It starts with 19 and contians exactly 9 digits.

Example: 192227101

2. course code

^[A-Z]{3}[0-9]{2}$

It contains 3 capital letters and 2 digits.

Example: DSA03

3. Date

^(0[1-9]|[12][0-9]|3[01])[-/](0[1-9]|1[0-2])[-/][0-9]{4}$

Example: 05-11-2026

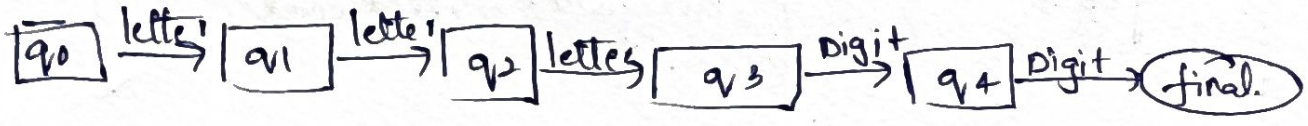
4. Email

^[A-Za-z0-9._%+-]+@saveetha|.com$

Example: nlp.dept@saveetha.com.

(b) DFA For course code

Course code: 3 capital letters followed by 2 digits



Transition table:

State	input	Next state
q_0	q_1	Letter
q_1	q_2	letter
q_2	q_3	Letter
q_3	q_4	Letter
q_4	q_4	Letter
q_5	q_4	digit
q_6	q_4	digit.

Trace:-

→ DSA03 → Accepted

→ CS503 → Rejected

→ DSA035 → Rejected

Q2) Morphological parser and porter stemmer

(a) Marphones.

word	Marphones	Type
unhappiness	un+happy+ness	Derivational
Foxes	fox +es	inflectional
cities	city+es	inflectional
Geese	goose	irregular inflectional.

walked walk +ed Inflectional

→ Derivational affix changes the meaning or word type

→ Inflectional affix shows tense, plural, etc.-

(b) Porter Stemmer

Word	Stem
Ponies	Poni
Motoring	Motor
Relational	relat-
Hopefulness	hope
Generalizations	general
Universities	Univers

over - Steaming!

over - Steaming means removing too many letters from a word

Example: Universe → Univers

Lemmatizer:

A lemmatizer gives the correct dictionary word

Examples:

→ Geese → Goose

→ walked → walk

→ Universities → University.

Q3) Query Normalization and Bigram model

(a) Query Normalization.

Query normalization converts words into their root forms.

word	Rootword	Affixtype	porter stem
Booked	Book	inflectional	Book
Booking	Book	inflectional	Book
classes	class	inflectional	class
Libraries	Library	inflectional	Librari

Analyzer and Stemmer disagreement:

- Analyzer: Libraries → Library
- Stemmer: Libraries → Librari

The search engine should use the analyzer because it gives a meaningful dictionary word.

(b) Bigram Language Model.

Formula:

$$p(w_i/w_{i-1}) = \frac{count(w_{i-1}, w_i)}{count(w_{i-1})}$$

1. probability of "i need a mom"

$$p(i/<s>) = 3/4$$

$P(need|i) = 2/3$

$P(a|need) = 1/2$

$P(room|a) = 1/2$

$$P = \frac{3}{4} \times \frac{2}{3} \times \frac{1}{2} \times \frac{1}{2}$$

$$\boxed{P = 0.125}$$

2. probability of "book a reciept"

$P(book|<s>) = 1/4$

$P(a|book) = 1/2$

$P(reciept|a) = 0$

$$\boxed{P = 0}$$

3. Add-one smoothing

Formula:

$$p = \frac{count + 1}{count(context) + V}$$

Q4) Predictive keyboard

(a) Stupid Backoff

Given $a = 0.4$

For book:

$P(book) = 0.50$

For the:

$P(the) = 0.4 \times 0.30 = 0.12$

For hall:

$$P(hall) = 0.60$$

Sentence probability:

$$P = 0.50 \times 0.120 \times 0.60$$

$$\boxed{P = 0.036}$$

(b) Linear interpolation

Formula:

$$P(w) = \lambda_1 P(w) + \lambda_2 P(w/previous) + \lambda_3 P(w/two\ previous)$$

Weight set A

$$(0.1, 0.3, 0.6)$$

word	probability
book	0.425
the	0.100
hall	0.392

Sentence probability:

$$P_A = 0.425 \times 0.100 \times 0.392$$

$$\boxed{P_A = 0.01666}$$

Sentence probability:

$$P_B = 0.360 \times 0.170 \times 0.234$$

$$\boxed{P_B = 0.0143208}$$

(c) Cross-Entry perplexity

Formula:

$$H = \frac{-1}{N} \log_2 p$$

$$PP = 2^H$$

Method	Cross-Entropy	perplexity
Stupid Backoff	1.594	3.028
Set A	1.969	3.915
Set B	2.042	4.118

Set A is better than set B as it has lower cross-entropy and perplexity.

(d) Entropy -

Given:

$$p = (0.5, 0.125, 0.125, 0.125)$$

formula:

$$H = -\sum p_i \log_2 p_i$$

$$\boxed{H = 1.625 \text{ bits}}$$

for 4 equally likely words:

$$H = \log_2 4$$

$$\boxed{H = 2 \text{ bits}}$$

Lower entropy means the keyboard is more confident in its prediction.

Q15) Hidden Markov Model (HMM) and Viterbi Algorithm.

Sentence: we can book

Tags:

→ PRP = pronoun

→ MD = Modal

→ VB = verb

→ NN = Noun.

(a) Viterbi Algorithm

Formula:

$$v_t(s) = \max [v_{t-1}(s') \times P(s|s') \times P(w|s)]$$

Step:1 calculate "can"

Tag	calculation	score
PRP	0.60 x 0.20	0.120
MD	0.05 x 0.60	0.025
VB	0.15 x 0.01	0.0015
NN	0.20 x 0.02	0.004

Step:2 calculate "book"

For VB:

0.12 x 0.45 x 0.05

0.0027

For NN:

$0.12 \times 0.10 \times 0.10$

0.0012

Viterbi table:-

Word	PRP	MD	VB	NN
can	0.120	0.025	0.0015	0.004
book	—	—	0.0024	0.0012

Best tag sequence:

PRP → VB

Probability:

0.0027

(b) Most frequent - Tagger

word	Most frequented tag
can	MD
book	VB

Output:-

can/MD book/VB

Viterbi output:

can/PRP book/VB

(c) Transformation Based Learning (Brill)

Rule:

Change can/MD to can/PRP when the next word is a verb

Baseline:

can/MD book/VB

After applying rule:

This rule improves tagging using context.

(d) Why HMM chooses VB for "book"

For VB:

$$P = 0.12 \times 0.45 \times 0.05$$

$$P = 0.0027$$

For NN:

$$P = 0.12 \times 0.10 \times 0.10$$

$$P = 0.0012$$

Since:

$$0.0027 > 0.0012$$

Conclusion:- HMM considers both transition and emission probabilities. Therefore, it selects the most probable tag sequence PRP → VB.